



KBase: Systems Biology Knowledgebase

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Company Name

“KBase integrates a variety of data and analysis tools, from DOE and other public resources, into an easy-to-use platform that leverages scalable computing infrastructure to perform sophisticated systems biology analyses.” -<https://www.kbase.us>

Developed by the Department of Energy Systems

Developer extensible platform for additional support

KBase Highlights

1

Internal databases

- RefSeq4
- Phytozome5,
- Biolog
- KEGG7
- BIGG8
- MetaCyc9

2

App Software Development Kits

- Users can develop their own apps
- Use apps developed by others with personal code

3

Easy-to-share

- Web-browser UI is easy to interact with
- Narratives and apps contain unique DOI for sharing
- Freely available for use

4

Narratives

- Functions like Jupyter notebook
- Easy setup and step-by-step analysis

Use Cases

Read Processing

Read Libraries
Read
creation/splitting
File Inputs:
SRA, FASTQ,
Compressed files

Genome Assembly & Annotation

Bin creation &
modification
Assemble Contigs
Genome Quality
Assessments &
Annotations
RASTtk, Prokka,
DRAM

Sequence Analysis

Blast
Multiple sequence
alignment
HMMER,
GBLOCKS,
Qualimap

Metabolic Modeling

Compare
metabolic
annotations
Build metabolic
models
Model fitness

Comparative Genomics

Compute/compare
genomes from
pangenome
Sequence
alignments

Microbial Community Interactions

Annotate
metagenome
assemblies
Optimizing
Contigs



Example 1: Genome Extraction from Shotgun Metagenome Sequence Data



Developed by Dylan Chivian (DCChivian@lbl.gov),
Mikayla Clark (clarkmm1@ornl.gov), and Sean
Jungbluth (sjungbluth@lbl.gov)

Objectives

Measure Taxonomic Population Structure

Perform quality control of reads

Assemble population structure based on Environmental shotgun reads

Assemble Metagenomes

Compare metagenome quality through contigs
Assess bin quality

Extracting individual assemblies

Work with Metagenome-assembled Genomes (MAGs)

Binning and assessing MAG quality

Gene annotation

Placing MAGs in reference species trees

Dataset: Compost Enrichment Metagenome (37A), as noted in

Wu YW, Higgins B, Yu C, Reddy AP, Ceballos S, Joh LD, Simmons BA, Singer SW, VanderGheynst JS. Ionic Liquids Impact the Bioenergy Feedstock-Degrading Microbiome and Transcription of Enzymes Relevant to Polysaccharide Hydrolysis. mSystems. 2016 Dec 13;1(6):e00120-16. doi: 10.1128/mSystems.00120-16. PMID: 27981239; PMCID: PMC5155067.

Step 1: Read Hygiene

Import FASTQ/SRA files (Illumina format)

Assess Read Quality Using FastQC

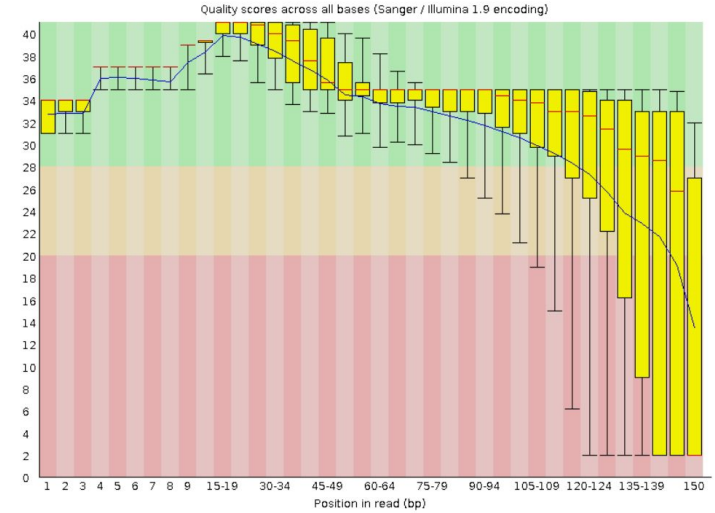
Results:

- Sequence Quality (per base, per tile, etc.)
- Sequence Content (GC, N)
- Other sequence distributions (length, duplication, overrepresentation)

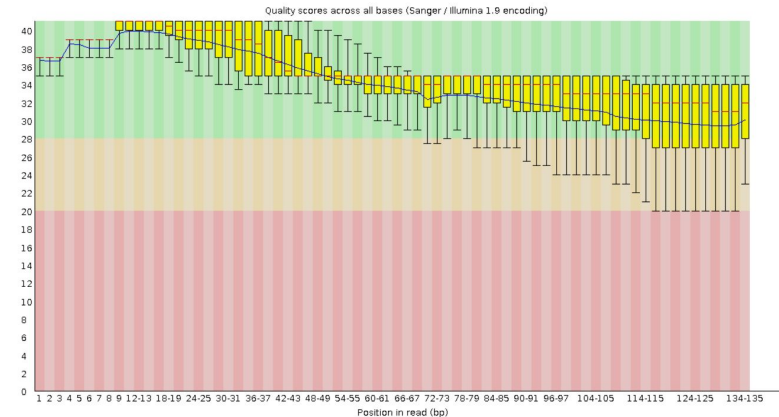
Trim Reads using Trimmomatic

- Trim paired- or single-end Illumina reads
- Can improve sequence quality
- Output PairedEndLibrary

✖ Per base sequence quality



✔ Per base sequence quality

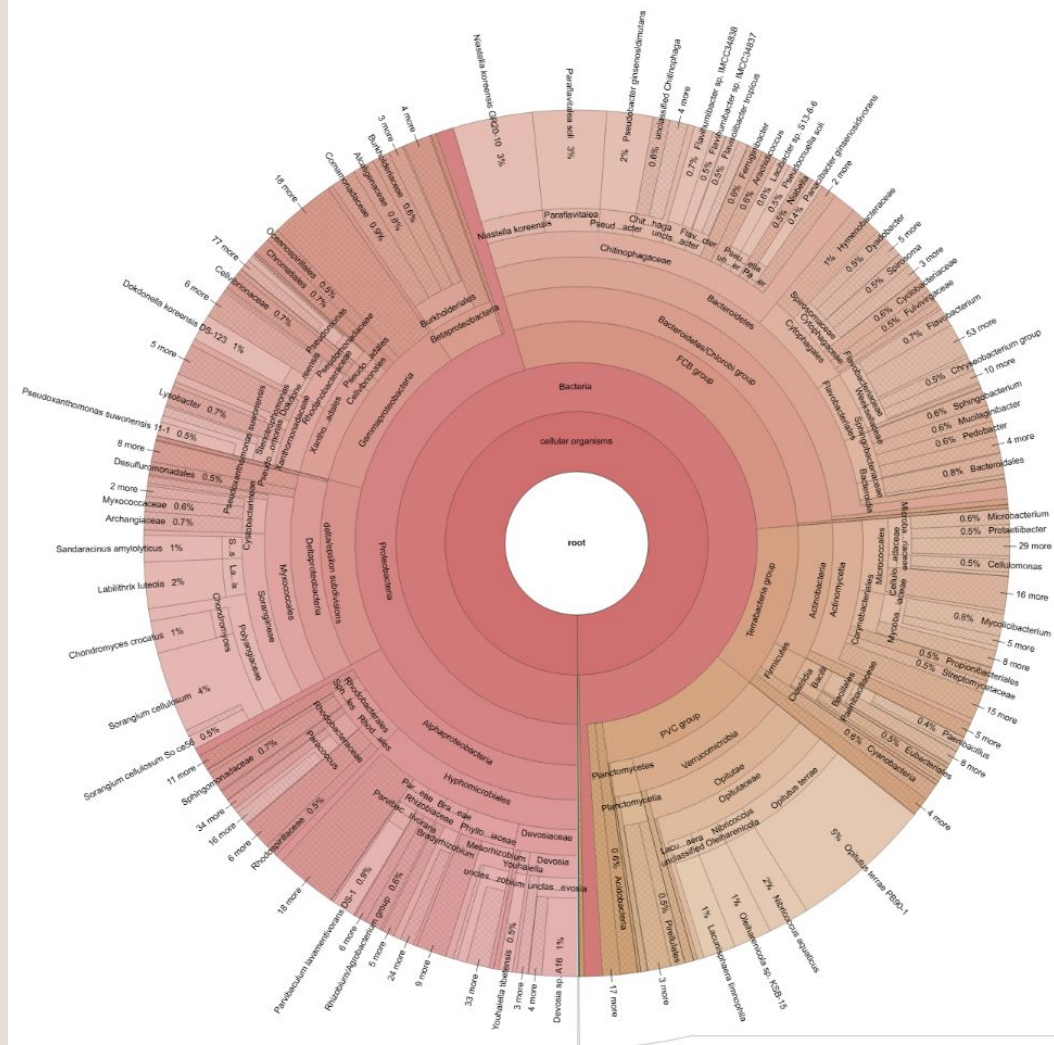


Step 2: Taxonomic Classification

Import SingleEndLibrary

Use Kaiju to classify results

- Kaiju
 - Program for sensitive taxonomic classification of high-throughput sequencing reads
 - Protein level classification allows for better performance vs. nucleotide
- Results:
 - Classification display
 - PairedEndLibrary

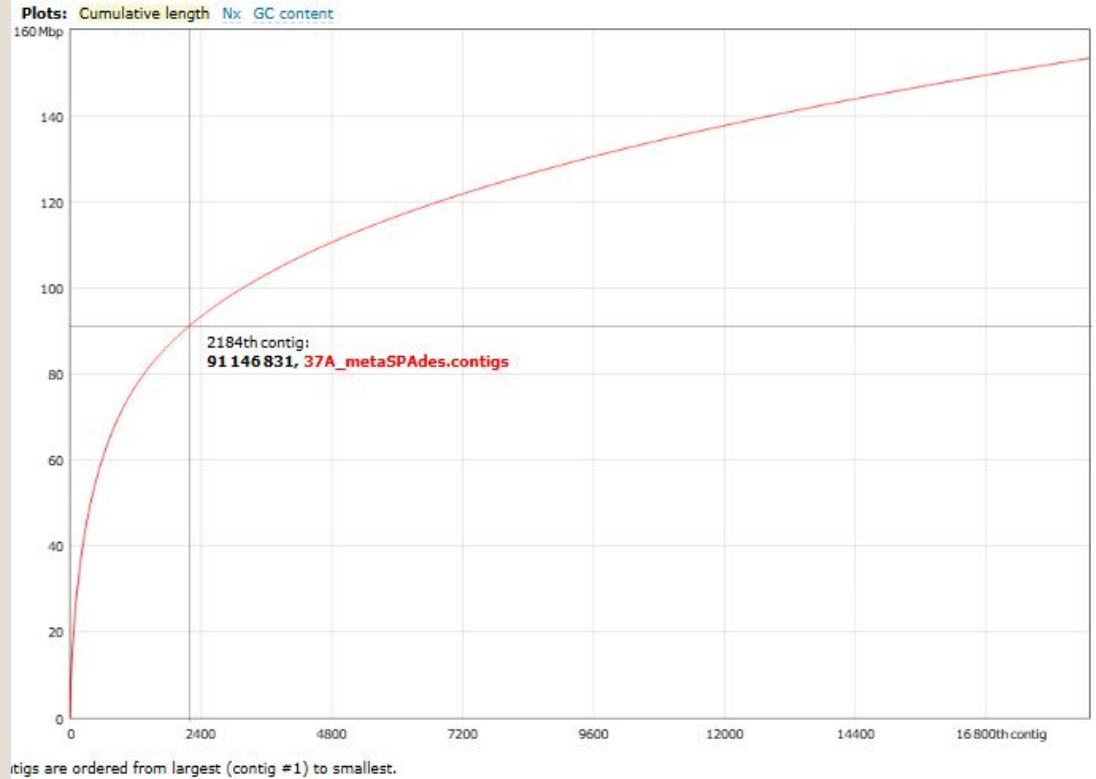


Step 3: Read Assembly

Import PairedEndLibrary

Use one of three assemblers

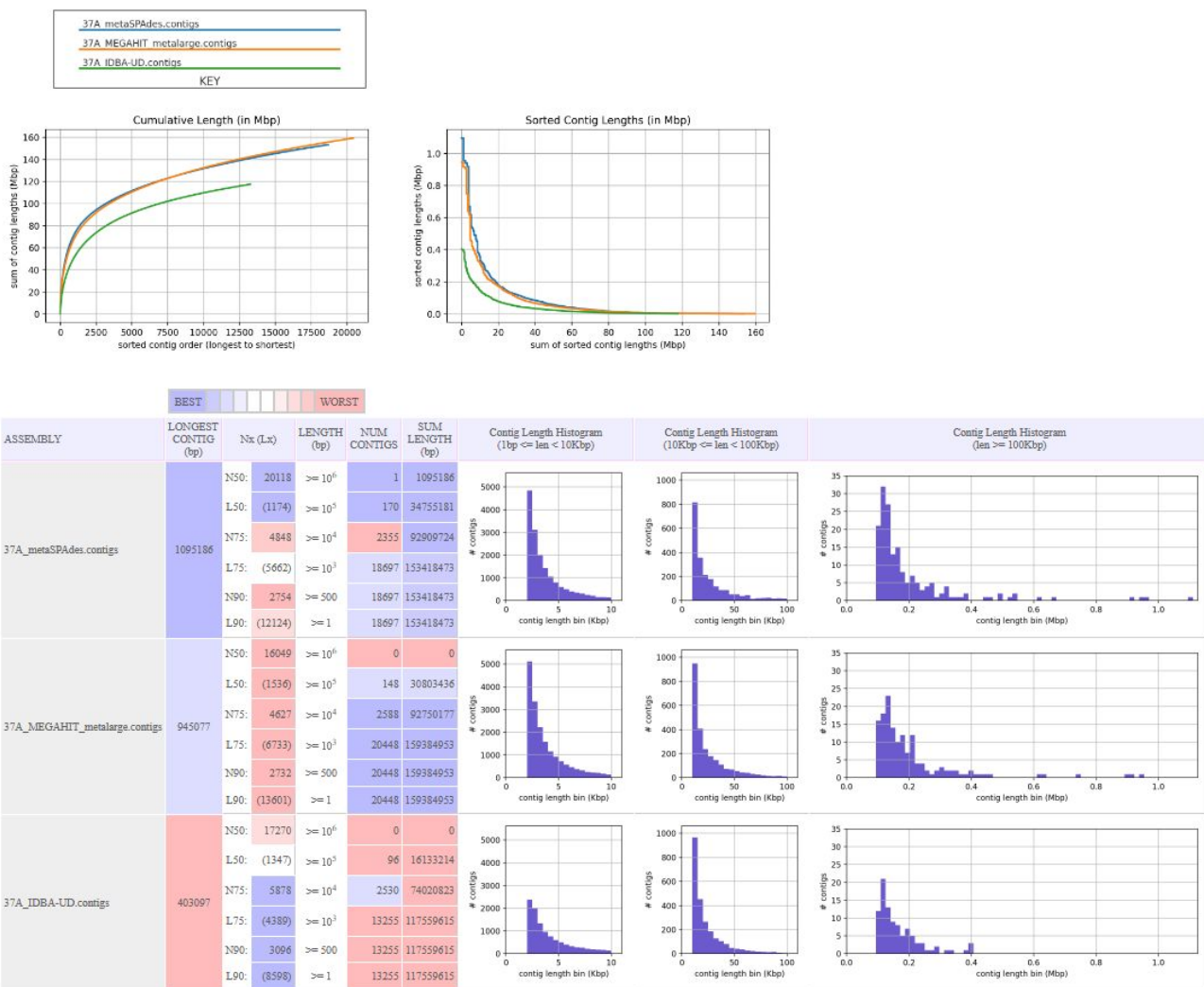
- MetaSPADES
 - De Bruijn graph of all reads using SPAdes
- MEGAHIT
 - Uses succinct de Bruijn graphs with a multiple kmer strategy
 - Faster
 - Memory efficient
- IDBA-UD
 - Iterative graph-based assembler
 - Based on IDBA algorithm
 - Good for data of highly uneven sequencing depth
 - Small to large k



Step 4: Compare Contigs

Custom Kbase comparison
function

Similar functionality to QUAST



Step 5: Contig Binning

Import contig Assemblies and reads file

Use one of three assemblers

- CONCOCT
 - Gaussian mixture models for contig cluster
 - Based on sequence composition
- MaxBin2
 - Expectation-Maximization algorithm
 - Based on tetranucleotide frequencies of contigs
- Metabat2
 - Calculates probabilistic distances based on tetranucleotide frequency (TNF) and abundance
 - K-medoid clustering

Binned contigs: 18682

Input contigs: 18697

Number of bins: 70

Bins: 41

Input Contigs: 18697

Binned Contigs: 15893 (85.0%)

Unbinned Contigs: 2804 (15.0%)

Contigs Too Short: 0 (0.0%)

Summed Length of Binned Contigs: 136464507 (88.9%)

Summed Length of Unbinned Contigs: 16953966 (11.1%)

Summed Length of Short Contigs: 0 (0.0%)

Binned contigs: 15974

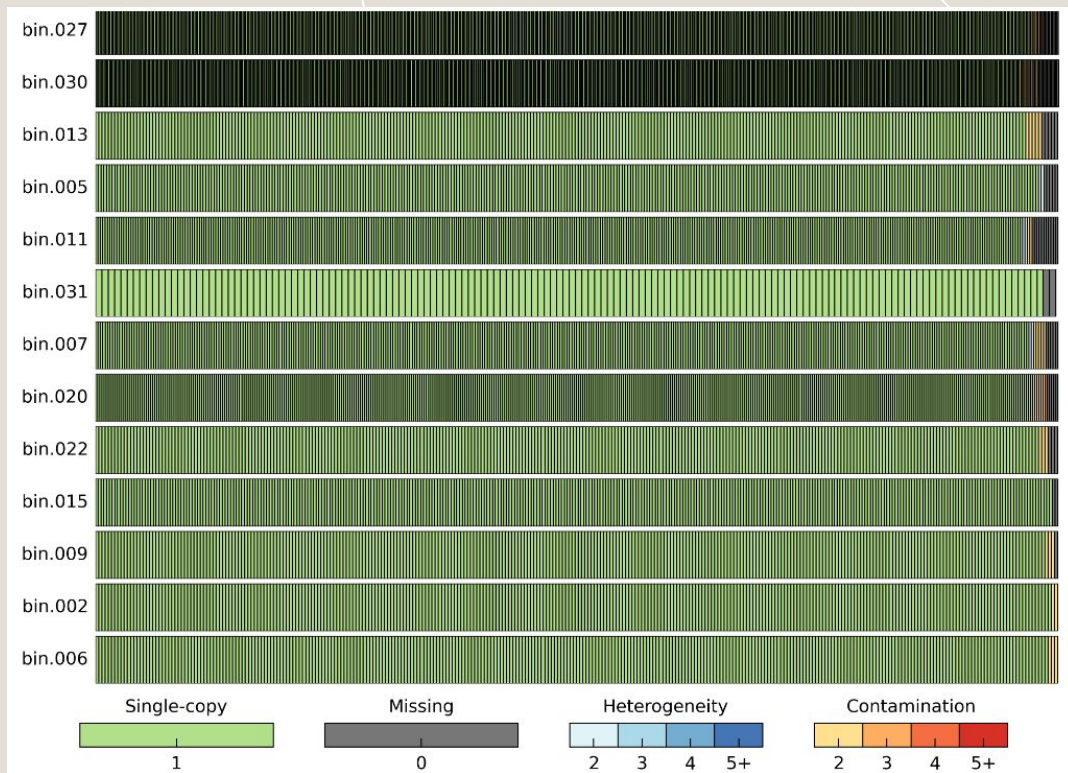
Input contigs: 18697

Number of bins: 48

Step 6: Optimize Contigs

Import BinnedContigs

- DAS Tool
 - Optimize contigs based on multiple inputs
 - Dereplication, aggregation and scoring strategy
- CheckM
 - Lineage-specific collocated marker sets
 - Assess genome quality
 - Filter bins

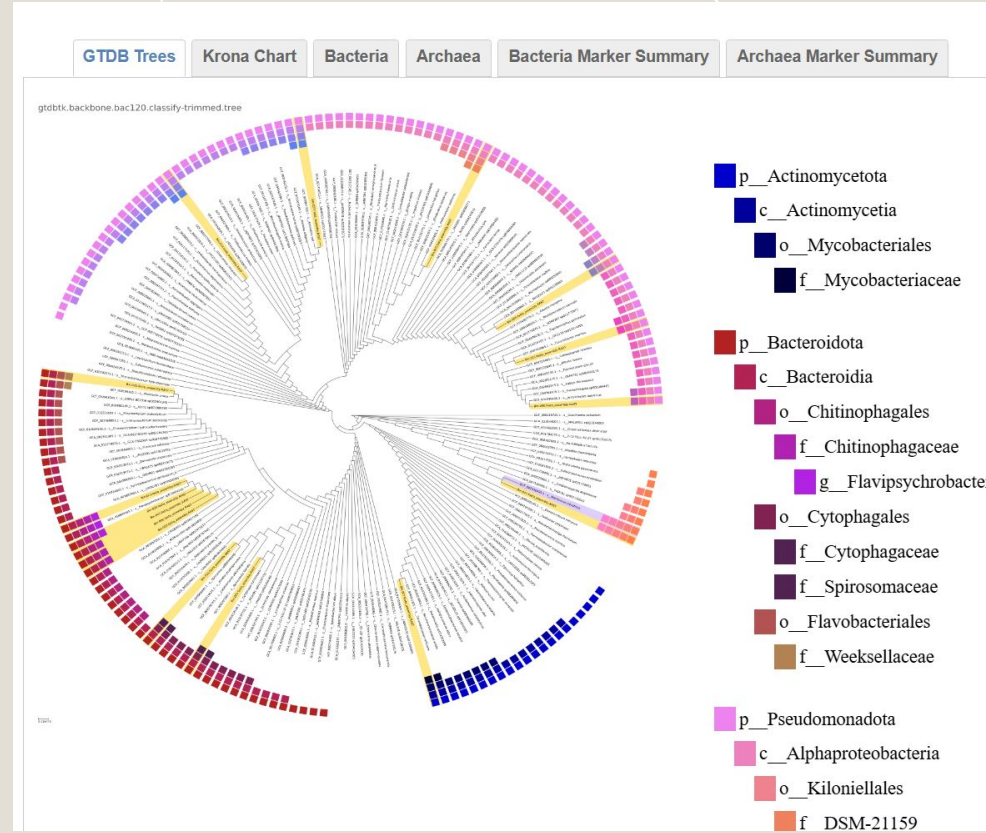


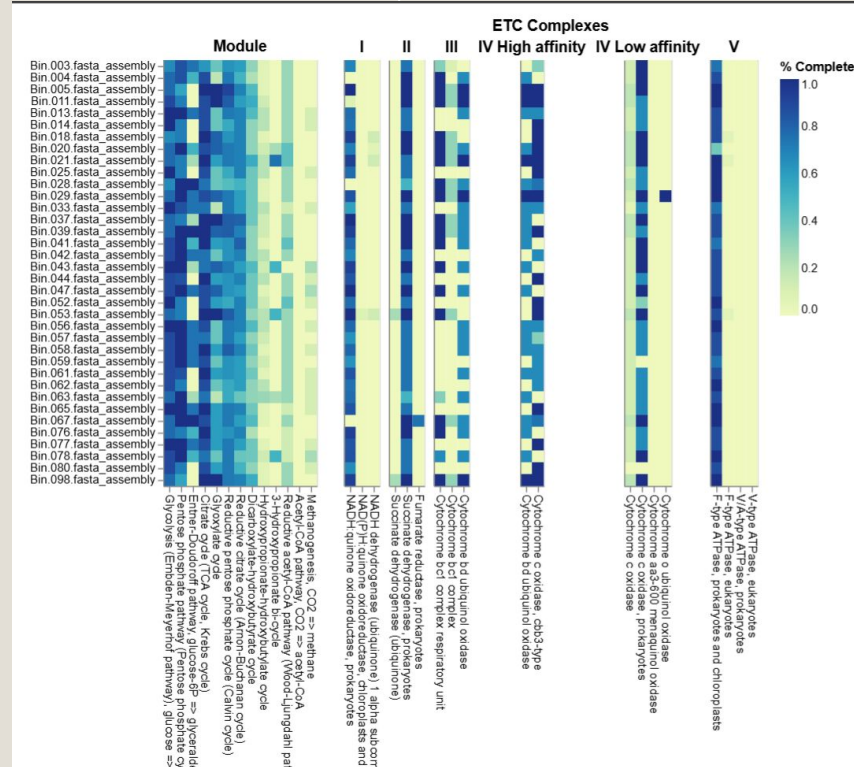
Step 7: Genome Annotation

Import BinnedContigs and extract individual bins

Annotate genomes through RASTtk

- Rapid Annotations using Subsystems Technology tool kit
- Compares genomes, proteins, and user annotations with databases to create annotations

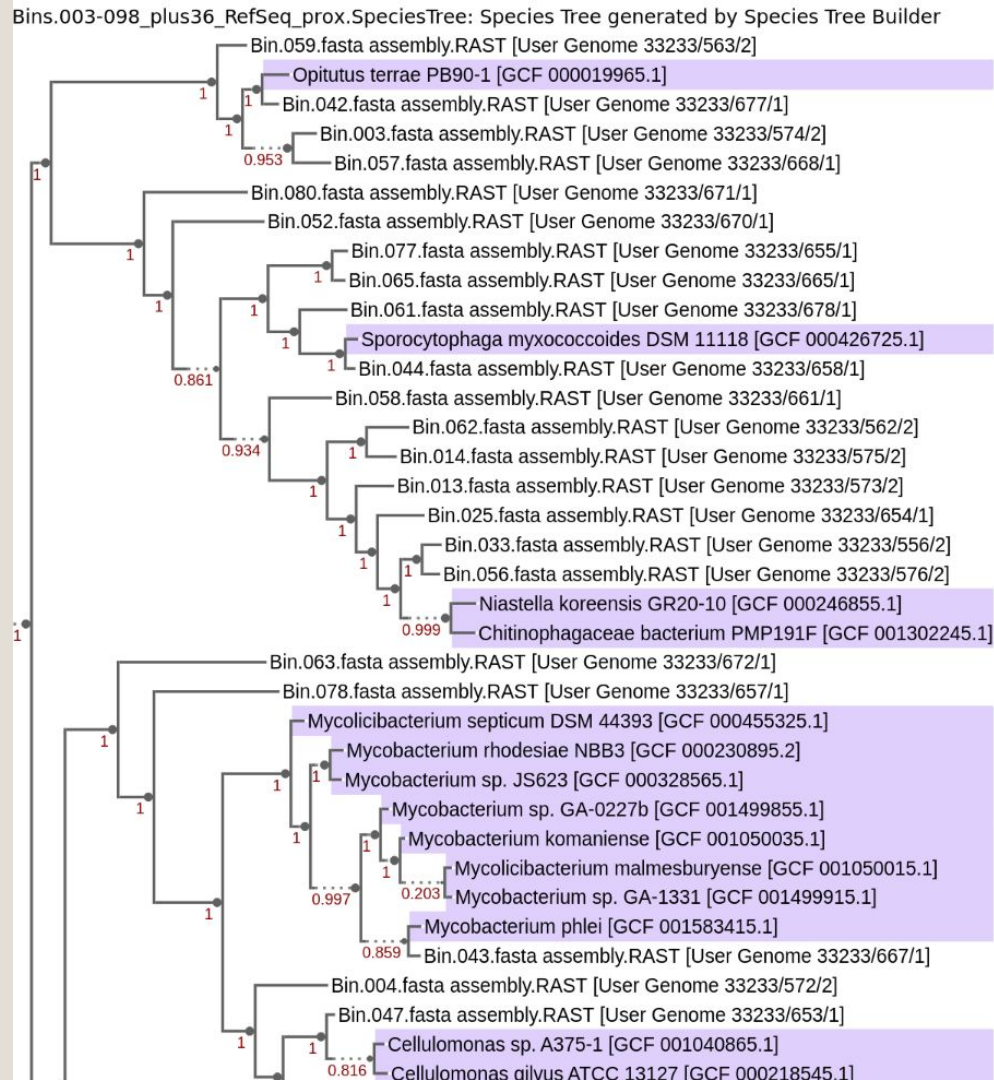


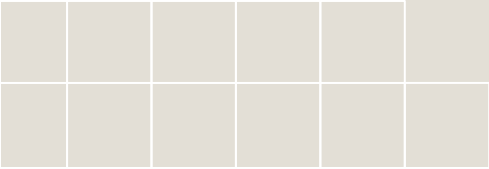


Additional Work: Species Trees

Can create SpeciesTrees based on created GenomeSet files

Manually insert species into tree using GenomeSet files





Example 2: Genome Extraction from Shotgun Metagenome Sequence Data

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Dataset & Tools

What I Used And Where it Came From

Source: Zenodo (public dataset), licensed CC-BY 4.0

Files:
CAMISIM_Illumina_R1.anonymous.G0_T0.fq.gz and
CAMISIM_Illumina_R2.anonymous.G0_T0.fq.gz

What I Changed Before Import (Pre-processing)

KBase import failed because the FASTQ had IUPAC ambiguous letters (e.g., R,Y,S,W,K,M,B,D,H,V) instead of just A,C,G,T,N.

KBase wants certainty: A/C/G/T or a placeholder N. I swapped all the “maybes” to N so the importer would accept the files.

KBase objects created (inputs you see in the Narrative)

Use Import from Staging → FASTQ Reads
NonInterleaved with both cleaned files.

Assemble Reads with
MEGAHIT v1.2.9

Dataset: Krakau, S. (2021). CAMISIM hybrid time series data (v1.0.0). Zenodo. DOI: 10.5281/zenodo.5155395
Software: MEGAHIT v1.2.9; QUAST (KBase integrated report).
License: CC-BY 4.0 (dataset).

Provenance summary:

I downloaded the CAMISIM G0_T0 paired-end Illumina reads from Zenodo (DOI: 10.5281/zenodo.5155395). The importer first failed because the FASTQ contained IUPAC ambiguity codes; I replaced those with N and re-compressed, then re-imported as CAMISIM_G0_T0_reads_clean. I assembled with MEGAHIT v1.2.9 (preset meta-sensitive, min contig 2,000 bp) producing MEGAHIT_assembly. Quality metrics came from the integrated QUAST report. These steps and names match my KBase Narrative, so the analysis is fully reproducible.

```
abdullahdawud-sulaiman@Abdullahs-MacBook-Pro Downloads % gunzip CAMISIM_Illumina_R1.anonymous.G0_T0.fq.gz
gunzip CAMISIM_Illumina_R2.anonymous.G0_T0.fq.gz

abdullahdawud-sulaiman@Abdullahs-MacBook-Pro Downloads % awk 'NR%4==2 {gsub(/[RYSWKMBDHV]/,"N")}' \
CAMISIM_Illumina_R1.anonymous.G0_T0.fq \
> CAMISIM_Illumina_R1.G0_T0.cleaned.fq

awk 'NR%4==2 {gsub(/[RYSWKMBDHV]/,"N")}' \
CAMISIM_Illumina_R2.anonymous.G0_T0.fq \
> CAMISIM_Illumina_R2.G0_T0.cleaned.fq

abdullahdawud-sulaiman@Abdullahs-MacBook-Pro Downloads % gzip CAMISIM_Illumina_R1.G0_T0.cleaned.fq
gzip CAMISIM_Illumina_R2.G0_T0.cleaned.fq
```

Import from Staging Area
Import files into your Narrative as data objects

Reset batch job finished with **success: 1 success** [View Configure](#) [Info](#) [Job Status](#) [Result](#)

Data type
FASTQ Reads NonInterleaved

File Paths [CREATE IMPORT TEMPLATE](#)

Forward/left FASTQ file path: CAMISIM_Illumina_R1.G0_T0.cleaned.fq.gz
Reverse/right FASTQ file path: CAMISIM_Illumina_R2.G0_T0.cleaned.fq.gz

1
Reads object name: CAMISIM_G0_T0_reads_clean

Parameters (3 advanced parameters hidden) [show advanced](#)

Sequencing technology: Illumina
Single genome: ☐

Step 1: Read Hygiene

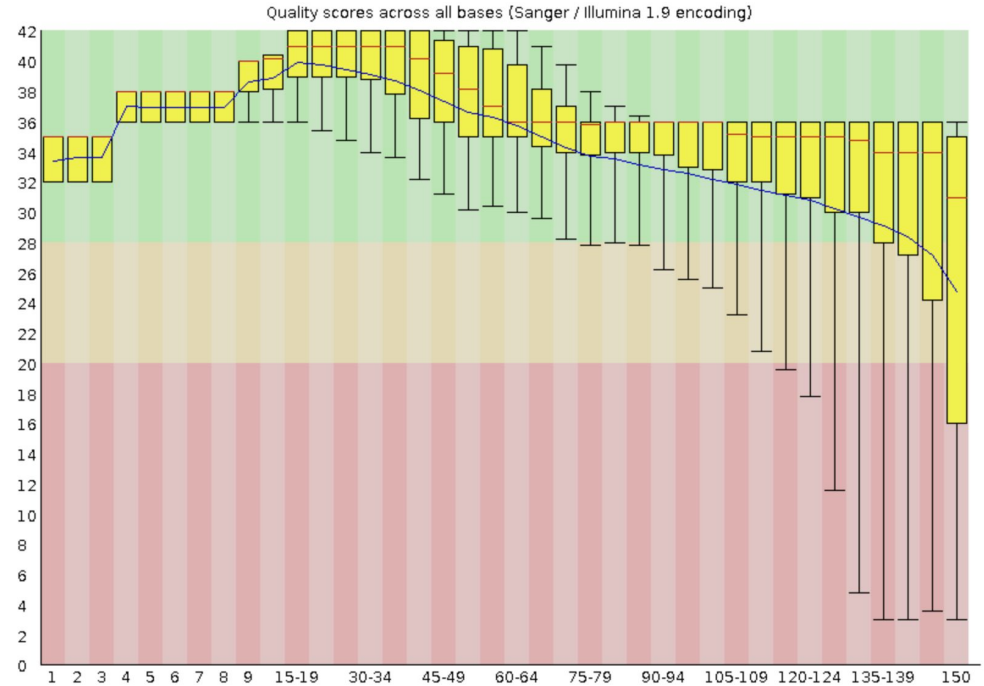
Import FASTQ files (Illumina paired-end)

Assess Read Quality Using FastQC

- Ran FastQC on my cleaned paired-end reads
CAMISIM_GO_T0_reads_clean
- The Per-base sequence quality plot (right) shows:
 - Bases 1–100 are in the green zone (high confidence).
 - Quality slowly drops toward the end of the reads
- Overall, the reads are high quality and suitable for assembly.

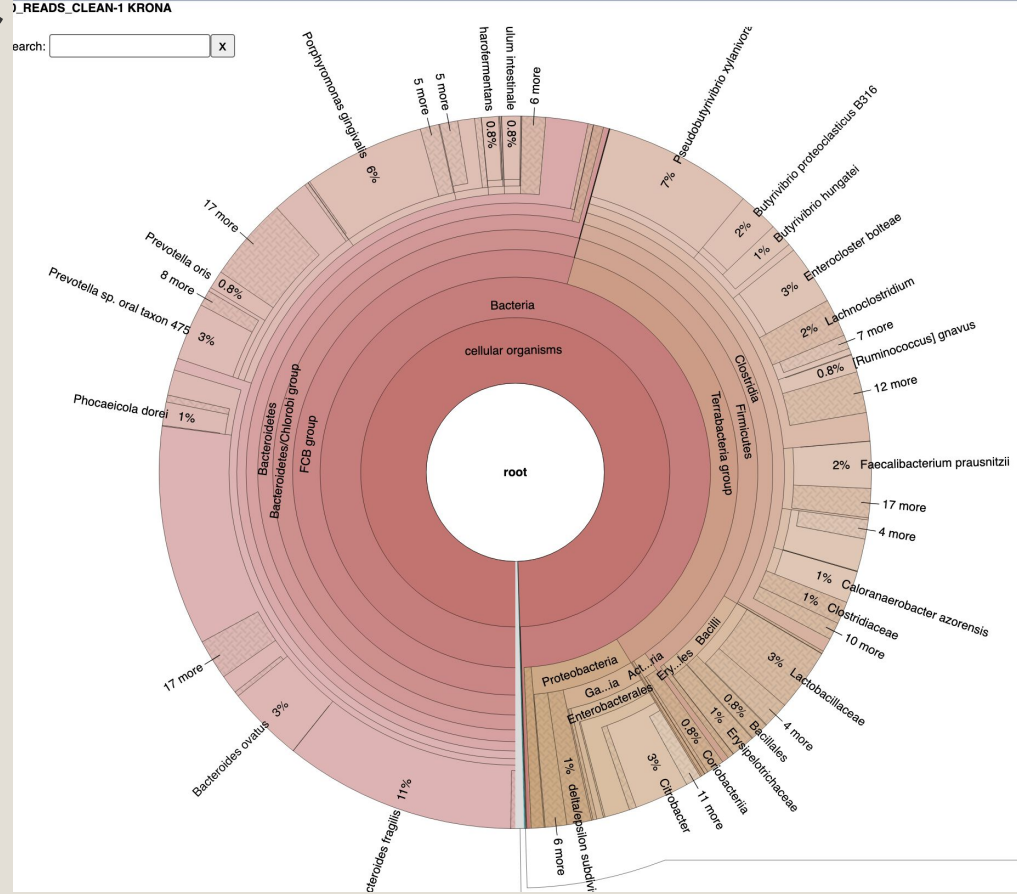


Per base sequence quality



Step 2: Taxonomic Classification

- Kaiju
 - Program for taxonomic classification of metagenomic reads.
 - Translates DNA reads into proteins and compares them to a reference protein database.
- Results:
 - Produced a Krona donut plot (right) showing the relative abundance of taxa
 - Most classified reads belong to Bacteria (Bacteroidetes, Firmicutes)
 - Smaller fractions come from Proteobacteria and other minor groups.



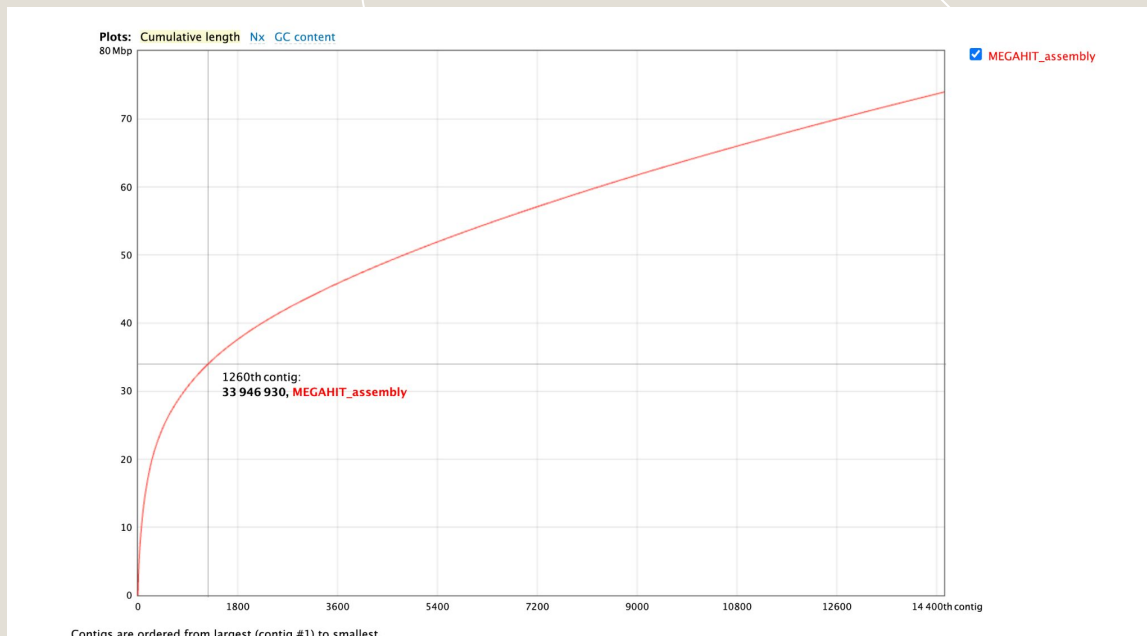
Step 3: Read Assembly

Import PairedEndLibrary

- Input: cleaned paired-end reads
CAMISIM_GO_TO_reads_clean
from Step 1.

Using MEGAHIT assembler

- MEGAHIT
 - Memory-efficient succinct de Bruijn graph and multiple k-mer sizes.
 - Designed for large, complex metagenomes; faster and more memory-friendly.
 - Output assembly: MEGAHIT_assembly



Step 3: Read Assembly

Assembly Result

- Number of contigs ≥ 1000 bp:
14,542
- Largest contig: 488,566 bp
- Total assembly length: 73,939,891 bp (~73.9 Mb)
- N50: 6,201 bp

Statistics without reference MEGAHIT_assembly

# contigs	14 542
# contigs (≥ 0 bp)	14 542
# contigs (≥ 1000 bp)	14 542
# contigs (≥ 10000 bp)	956
# contigs (≥ 100000 bp)	51
# contigs (≥ 1000000 bp)	0
Largest contig	488 566
Total length	73 939 891
Total length (≥ 0 bp)	73 939 891
Total length (≥ 1000 bp)	73 939 891
Total length (≥ 10000 bp)	31 263 031
Total length (≥ 100000 bp)	8 864 634
Total length (≥ 1000000 bp)	0
N50	6201
N75	2805
L50	1695
L75	6607
GC (%)	45.26

Mismatches

# N's	0
# N's per 100 kbp	0

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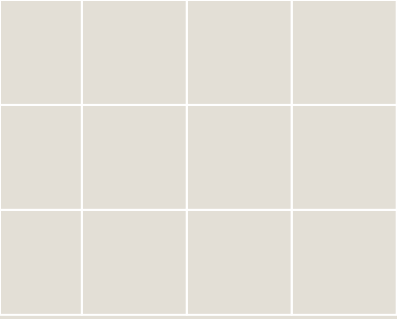
Parks DH, Imelfort M, Skennerton CT, Hugenholtz P, Tyson GW. CheckM: assessing the quality of microbial genomes recovered from isolates, single cells, and metagenomes. *Genome Research*. 2015;25: 1043–1055. doi:10.1101/gr.186072.114 <http://genome.cshlp.org/content/25/7/1043.long>

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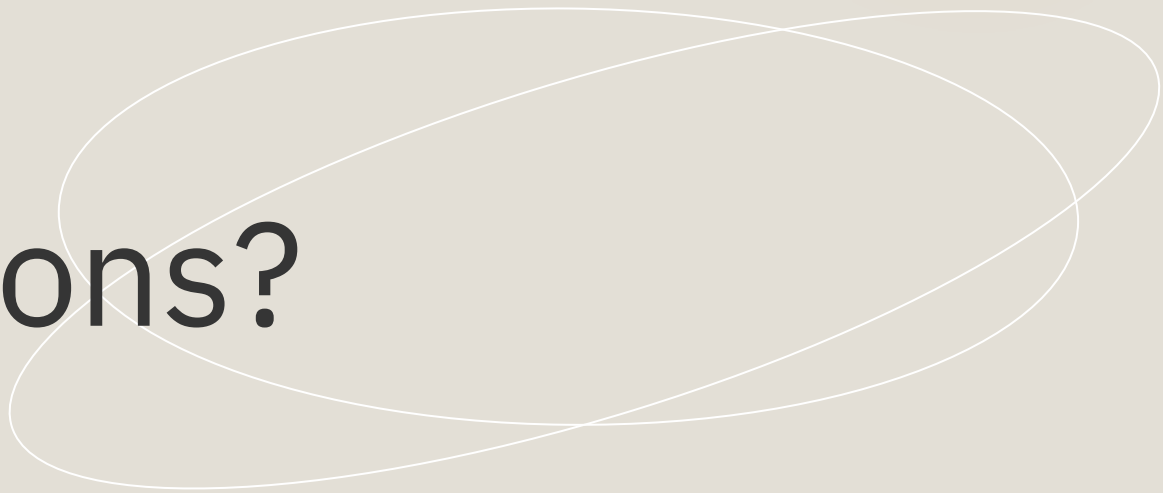
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Any questions?
Ask away!





Thank you